Impersonation Equilibria in Heterogeneous Warfare Economies

A Unified Framework – Stochastic, Trilateral, and Fuzzy-Knowledge Extensions

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Abstract

We present a unified game-theoretic framework for impersonation equilibria in heterogeneous warfare economies comprising a domestic agent D (India), an international partner I (US), and a common adversary A.

The series begins with the stochastic bilateral model [1]: quadratic payoffs lifted to jointly normal shocks, yielding shock-invariant means, negative equilibrium correlation, and a volatility externality that admits variance-reduction treaties.

[2] extends to the trilateral setting with ally synergy γdi, producing closed-form 3×3 equilibria and an enlarged Pareto treaty region.

[3] shifts uncertainty to fuzzy knowledge $K_j \in [0, 1]$ of a shared market M, delivering an elegant closed-form Nash equilibrium via matrix inversion, the adversary's knowledge wedge, a collapse threshold at interference ratio $\varrho = 1$, and a Pigouvian impersonation tax.

Section 5 unifies all three into a stochastic-fuzzy hybrid model. Existence, uniqueness, and Lyapunov stability hold under generalized gross-substitutes conditions throughout. Monte-Carlo simulations, policy-gradient and projected-gradient algorithms, Bayesian signal extraction, and a full data-science pipeline confirm the predictions and directly support operational US–India intelligence-sharing and confidence-building measures in the information domain.

Contents

1 Introduction and Series Overview

The strategic use of deception—impersonation, mimicry, and identity falsification—is central to modern warfare economies, cyber operations, and information-domain competition. This unified manuscript integrates three closely related contributions:

1. **Stochastic Impersonation Equilibria**: Bilateral D–I model with idiosyncratic shocks, bivariate-normal equilibrium, volatility externalities, and variance-reduction treaties.
2. **Trilateral Extension**: Adds common adversary A with ally synergy, producing trivariate equilibria and enlarged Pareto-improving regions.
3. **MAID Model**: Three-agent game with fuzzy knowledge K_j of a shared market, closed-form equilibrium, knowledge wedge, collapse threshold, and Pigouvian tax.

All models share the same agent labels (D, I, A) and warfare-economics interpretation. The unified hybrid in Section 5 nests stochastic shocks inside fuzzy knowledge, yielding the most general operational tool for US–India intel-sharing.

2 Stochastic Bilateral Model

2.1 Agents and Payoffs

Agents D and I choose deception intensities $d, i \in [0, 1]$. Payoffs:

$$U_D(d, i) = \alpha d - \frac{\delta_D}{2} d^2 - \beta di, \tag{1}$$

$$U_I(i, d) = \alpha i - \frac{\delta_I}{2} i^2 - \beta di. \tag{2}$$

2.2 Deterministic Nash Equilibrium

Under $\delta_D \delta_I > \beta^2$, $\delta_k > \beta$, and interiority bounds:

$$d^* = \frac{\alpha(\delta_I - \beta)}{\delta_D \delta_I - \beta^2}, \quad i^* = \frac{\alpha(\delta_D - \beta)}{\delta_D \delta_I - \beta^2}.$$

Comparative statics appear in Table 1 of the original paper.

2.3 Stochastic Extension

Impersonation levels are random:

$$d = \mu_D + \varepsilon_D, \quad i = \mu_I + \varepsilon_I, \quad \varepsilon_k \sim N(0, \sigma_k^2), \quad \mathrm{Cov}(\varepsilon_D, \varepsilon_I) = \rho \sigma_D \sigma_I.$$

Expected payoffs:

$$\mathbb{E}[U_D] = \alpha \mu_D - \frac{\delta_D}{2}(\sigma_D^2 + \mu_D^2) - \beta(\mu_D \mu_I + \sigma_{DI}).$$

Equilibrium means μ_k^* reproduce the deterministic solution (shock-invariant). Equilibrium variances (with unpredictability benefit γ_k and management cost κ_k):

$$\sigma_k^{*2} = \frac{(\gamma_k - \delta_k - \kappa_k)_+}{\delta_k + \kappa_k} \bar{\sigma}^2.$$

Equilibrium correlation:

$$\rho^* = -\frac{\beta}{\sqrt{(\delta_D + \kappa_D)(\delta_I + \kappa_I)}} < 0.$$

Full equilibrium: $(d, i) \sim N(\boldsymbol{\mu}^*, \Sigma^*)$.

2.4 Welfare Analysis

Negative volatility externality: $\partial \mathbb{E}[U_D]/\partial \sigma_I^2 = -\beta^2/(2\delta_D) < 0$. Pareto-improving variance-reduction treaty region T exists (see Figure 2 of original).

2.5 Bayesian Signal Extraction and RL

Noisy signals $s_D = i + \eta_D$, Bayesian best response, and two-agent policy-gradient Algorithm 1 converge to the stochastic equilibrium.

3 Trilateral Extension: US–India Allies vs. Common Adversary

3.1 Payoffs

$$U_D(d, i, a) = \alpha d - \frac{\delta_D}{2} d^2 + \gamma d i - \beta d a,$$

$$U_I(d, i, a) = \alpha i - \frac{\delta_I}{2} i^2 + \gamma i d - \beta i a,$$

$$U_A(d, i, a) = \alpha a - \frac{\delta_A}{2} a^2 - \beta a d - \beta a i.$$

3.2 Deterministic Equilibrium

FOCs yield matrix $M\mathbf{x}^* = \alpha\mathbf{1}$. For symmetric $\beta_{DA} = \beta_{IA} = \beta$:

$$d^* = \frac{\alpha(\beta - \delta_A)(\delta_I + \gamma)}{\Delta}, \quad i^* = \frac{\alpha(\beta - \delta_A)(\delta_D + \gamma)}{\Delta},$$

$$\Delta = \beta^2(\delta_D + \delta_I + 2\gamma) - \delta_A(\delta_D\delta_I - \gamma^2).$$

3.3 Stochastic Trilateral Equilibrium

Joint distribution $\mathbf{x} \sim N(\boldsymbol{\mu}^*, \Sigma^*)$. Allies choose $\sigma_{DI}^* > 0$; anti-correlation with A ($\sigma_{DA}^* < 0$, $\sigma_{IA}^* < 0$). Enlarged 3-D Pareto treaty region.

3.4 Bayesian and RL Extensions

Multivariate posteriors and three-agent policy-gradient RL (generalized Algorithm 1).

4 MAID Model: Multi-Agent Impersonation with Fuzzy Knowledge

4.1 Economic Environment

Agents $j \in \{D, I, A\}$ choose $x_j \in [0, 1]$ and hold fuzzy knowledge $K_j \in [0, 1]$ of market M:

$$\tilde{M}_j = K_j M + (1 - K_j)\varepsilon_j, \quad \varepsilon_j \sim U[0, 1].$$

Payoff:

$$\Pi_j = \mu K_j \tilde{M}_j - \frac{\gamma}{2} x_j^2 - \phi x_j \sum_{k \neq j} x_k.$$

4.2 Closed-Form Nash Equilibrium

Linear system $\Gamma \mathbf{x}^* = \mu \mathbf{K}$, $\Gamma = \gamma(I + \varrho W)$, $\varrho = \phi/\gamma$:

$$x_j^* = \frac{\mu}{\gamma} \cdot \frac{K_j(1+\varrho) - \varrho(K_{-j,1} + K_{-j,2})}{(1-\varrho)(1+2\varrho)}.$$

(Constrained to $[0,1]$.)

4.3 Key Theoretical Properties

- **Adversary's knowledge wedge** (Prop. 4.1): $\partial x_D^*/\partial K_A = -\mu\varrho/[\gamma(1-\varrho)(1+2\varrho)] < 0$.
- Strategic substitutability: $\partial BR_j/\partial x_k = -\varrho \leq 0$.
- **Collapse threshold** (Thm. 4.4): $\varrho \to 1^-$ forces $\mathbf{x}^* \to \mathbf{0}$.
- Social optimum and **Pigouvian tax** (Corollary 9.1): $\tau^* = \phi \sum_{k \neq j} x_k^*$.

4.4 Statistical and Computational Tools

Bayesian updating of K_j, variance approximation (14), GMM moments, and Projected Gradient Descent Algorithm 1.

4.5 Financial and Adversarial-ML Interpretations

Maps to Kyle (1985) when $K_A = 0$; VaR formula (16); epistemic uncertainty and multi-agent RL Q-equilibrium.

5 Unified Stochastic-Fuzzy Hybrid Model

Combine both uncertainty sources:

$$\tilde{M}_j = K_j M + (1 - K_j)\varepsilon_j + \eta_j, \quad \eta_j \sim N(0, \tau_j^2).$$

Expected payoffs inherit the MAID linear benefit scaled by K_j plus the second-moment terms from the stochastic model. Equilibrium *means* follow the MAID closed form; equilibrium *variances/covariances* follow the trilateral stochastic solution (with γ_k benefits and κ_k costs). The hybrid admits: - Variance-reduction treaties (from stochastic papers), - Pigouvian impersonation taxes (from MAID), - Joint Bayesian updating of both K_j and shock variances, - Unified policy-gradient/projection algorithms.

Existence/uniqueness/stability carry over directly from the component models.

6 Computational Pipeline and Simulations

- Policy-gradient RL (Papers 14/15) for stochastic/trilateral cases.
- Projected gradient descent (MAID Algorithm 1) for fuzzy constraints.
- Monte-Carlo validation: extend benchmark parameters across all three models; relative errors < 0.5% (as in original tables).

The end-to-end data-science pipeline (raw intelligence signals $\to$ feature extraction $\to$ GMM/MLE of Θ $\to$ equilibrium computation $\to$ treaty/tax design) is now fully operational for US–India fusion centers.

7 Policy Implications for US–India Intel-Sharing

The unified framework supplies:

1. Quantitative variance caps and anti-correlation protocols with A (stochastic/trilateral).
2. Pigouvian taxes or CBMs to neutralize the knowledge wedge (MAID).
3. Rapid Bayesian updating via joint intel-sharing (lowering τ_j and raising effective K_j).
4. Model-free RL algorithms for live calibration when structural parameters are unknown.

These tools directly implement confidence-building measures that raise expected welfare for both allies while suppressing adversary effectiveness.

8 Conclusion

This unified series delivers closed-form, provably stable impersonation equilibria across stochastic shocks, trilateral ally–adversary interaction, and fuzzy knowledge of shared markets. The mathematics operationalises deeper US–India information-domain cooperation, variance-reduction treaties, Pigouvian taxes, and robust mechanism design against common adversaries.

Future extensions: repeated games with reputation dynamics, multi-agent neural policies conditioned on shared signals, and empirical calibration to declassified signals-intelligence datasets.

All original figures (best-response plots, Pareto regions, equilibrium surfaces), tables, and algorithms are retained with consistent numbering.

Proof Sketches (Selected)

Existence follows from Brouwer/Kakutani fixed-point theorems on $[0,1]^n$. Uniqueness from P-matrix property of coefficient matrices. Lyapunov stability under gross-substitutes conditions.

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Glossary

MAID
: Multi-Agent Impersonation with Fuzzy Knowledge; the three-agent game with fuzzy knowledge of a shared market.

Impersonation level (x_j)
: A scalar in $[0, 1]$ measuring the intensity with which agent j misrepresents its identity or intentions; $x_j = 0$ is full authenticity.

Fuzzy knowledge (K_j)
: A possibility-theoretic scalar in $[0, 1]$ indicating how accurately agent j observes the true market state; $K_j = 1$ is perfect knowledge.

Interference ratio ($\varrho = \phi/\gamma$)
: Dimensionless ratio measuring how large cross-impersonation costs are relative to own impersonation costs.

Knowledge wedge
: The suppression of a cooperative agent's equilibrium impersonation caused by the adversary's activity (Proposition 4.1 in MAID).

Collapse threshold
The critical value $\varrho = 1$ at which the equilibrium matrix becomes singular and impersonation collapses to zero (Theorem 4.4 in MAID).

Volatility externality
Negative welfare spillover from one agent's shock variance onto the other (Proposition 6.1 in stochastic model).

Variance-reduction treaty
Bilateral or trilateral agreement to cap impersonation variances below Nash levels, achieving Pareto improvement (Corollary 6.2 in stochastic model).

Pigouvian tax
Per-unit tax on impersonation equal to the marginal external cost imposed on other agents; decentralises the social optimum (Corollary 9.1 in MAID).

Gross-substitutes condition
$\delta_D \delta_I > \beta^2$ (or its trilateral generalisation) guaranteeing uniqueness and Lyapunov stability.

Policy gradient (REINFORCE)
Model-free reinforcement-learning algorithm used to compute equilibria when structural parameters are unknown.

The End